



25.06.2026

Tag 4

Von KI erstellt, von Menschen kontrolliert.

Guten Morgen!

- RBAC Roles → Azure  
Roles → EntraID  

Labs (skillable) — AZ-900
— AZ-104

- Policies → Azure

GitHub Repo <https://github.com/www42/a5>

AZ-104

AZ-104
Lab 2a, Lab 1

RBAC Azure Policies

Learning Paths:

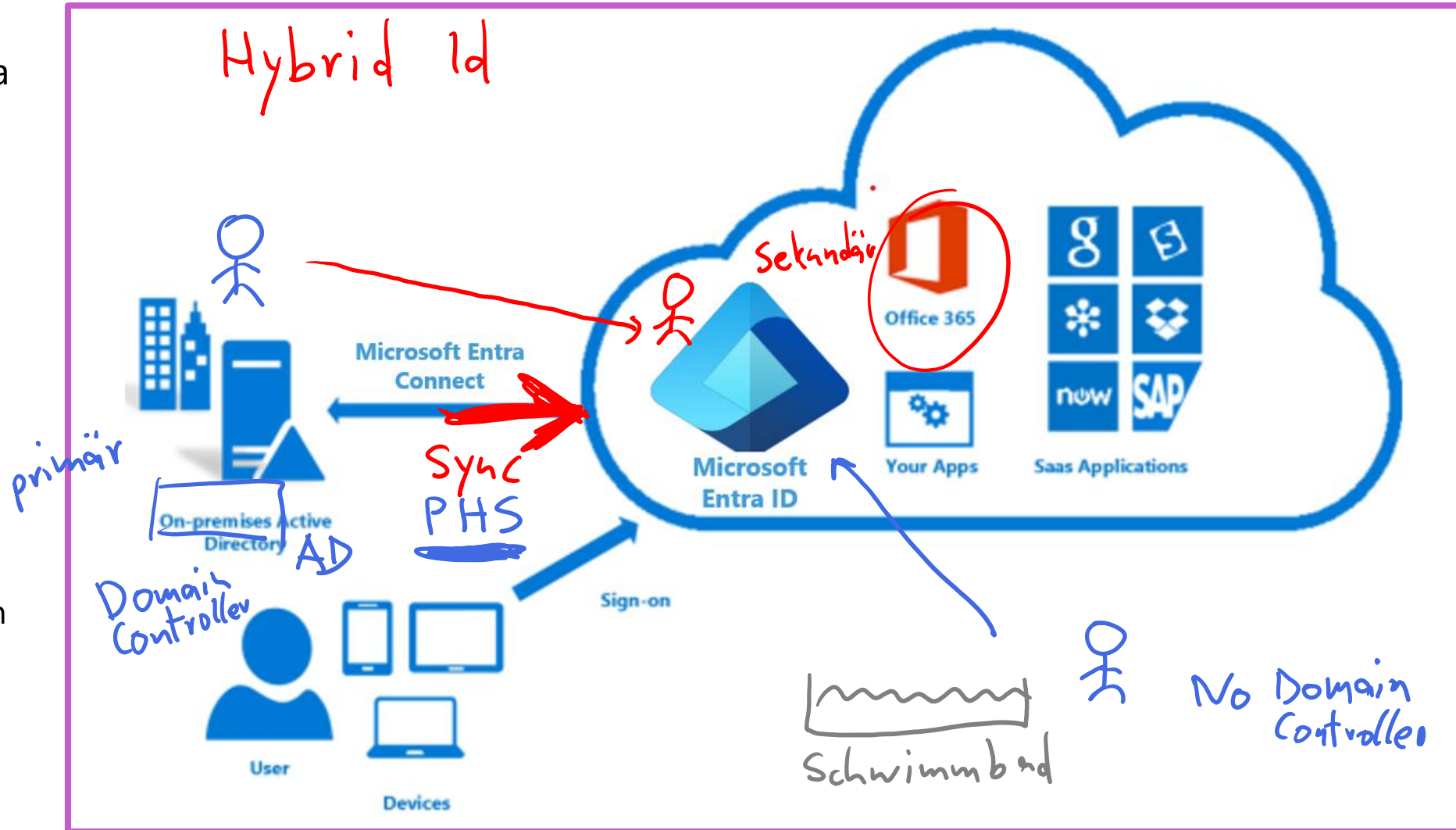
- Manage identities and governance in Azure ←
- Prerequisites for Azure administrators – CloudShell and templates
- Implement and manage storage in Azure
- Configure and manage virtual networks for Azure administrators
- Deploy and manage Azure compute resources
- Monitor and back up Azure resources

What happened yesterday

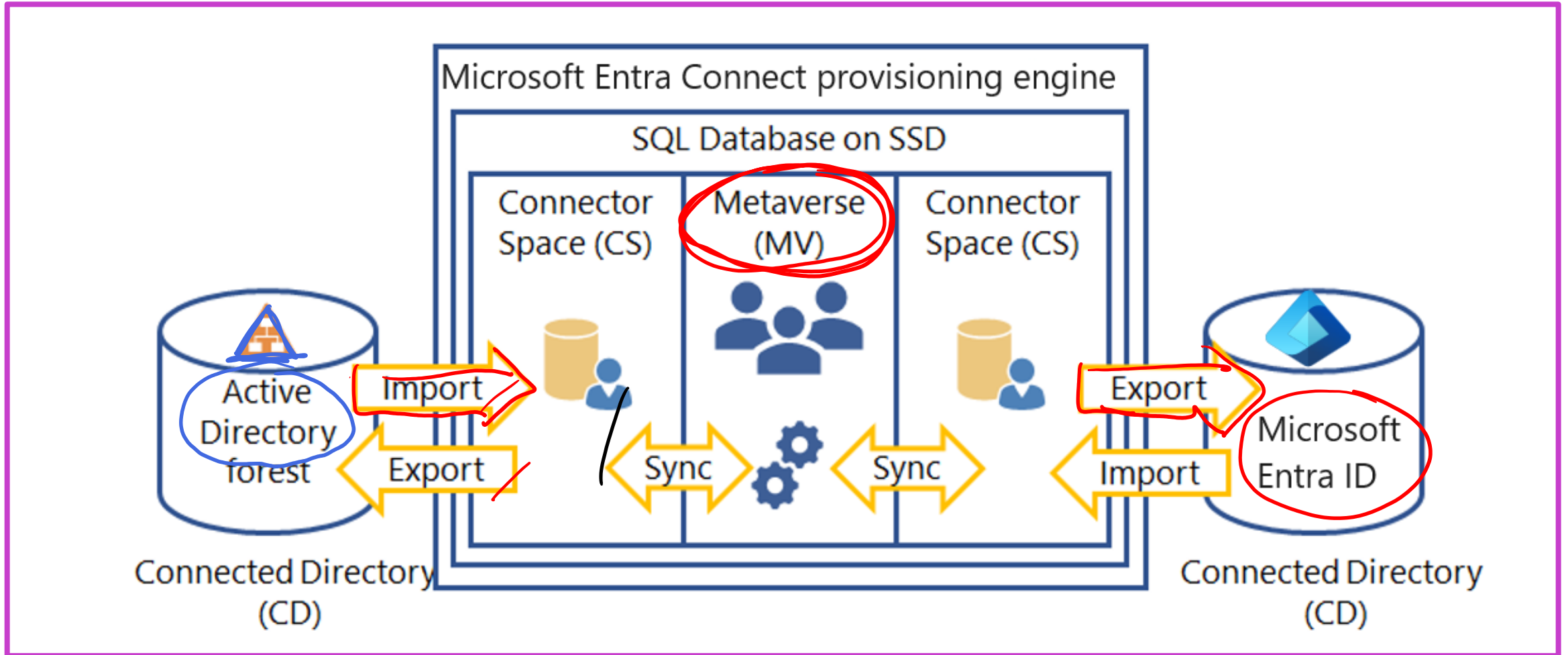


What is Microsoft Entra Connect?

- Microsoft Entra Connect is a solution that bridges an organization's on-premises Active Directory with your cloud-based Microsoft Entra ID
- Microsoft Entra Connect provides:
 - Synchronization
 - Password hash synchronization **PHS**
 - Pass-through authentication
 - Federation integration
 - Health monitoring



Microsoft Entra Connect component factors



TO DO

- Visual Studio Code (VS Code)
- Git / GitHub

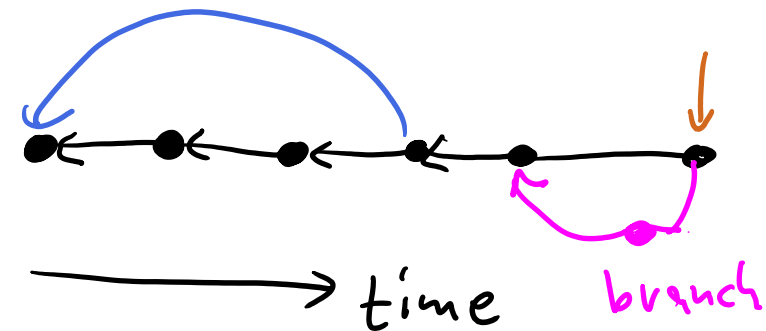
Headless state

- Vibe coding / AI
- Tag im Trainingcenter
- Prüfungsvorbereitung

ISE PowerShell

Editor + More

Versionskontrolle



AZ-104

Today

Learning Paths:

- Manage identities and governance in Azure ✓
- Prerequisites for Azure administrators – CloudShell and templates
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AZ-104

➔ Manage identities and governance in Azure



ACL

File Server

"Security"

Everyone

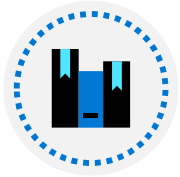
Full Control

Secure your Azure resources
with Azure role-based access
control (Azure RBAC)

Role
Based
Access
Control



Review Azure Resource Terminology



A **resource** is simply a single service instance in Azure



...



A **resource group** is a logical grouping of resources

Schublaade

Subscription



An **Azure Resource Manager template** is a JSON file that allows you to declaratively describe a set of resources



A **declarative syntax** is what a template uses to state what you intend to create

idempotent



A **resource provider** is service that supplies the resources you can deploy and manage through Resource Manager

Liefert Resource Type
Liefert Permission

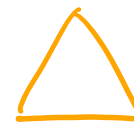
Microsoft.Compute/virtualMachines @ 2025-11-01

Implement Role-Based Access Control

Provides <u>fine-grained access management</u> of resources in Azure	Concepts
Built on Azure Resource Manager Segregate duties within your team Grant only the amount of access to users that they need to perform their jobs <i>Owner</i> <i>Rollen zuweisung</i>	Security principal. Object that represents something that is requesting access to resources Role definition. Collection of permissions that lists the operations that can be performed <i>json</i> Scope. Boundary for the level of access that is requested Assignment Attaching a role definition to a security principal at a particular scope: • Users can grant access described in a role definition by creating an assignment • Deny assignments are currently read-only and are set by Azure Blueprints and Azure Managed Apps

Azure

Entra ID



Compare Azure RBAC Roles to Entra ID Roles

RBAC roles provide fine-grained access management

Azure RBAC roles	Entra ID roles
Manage access to Azure resources	Manage access to Entra ID objects
Scope can be specified at multiple levels	Scope is at the tenant level
Role information can be accessed in the Azure portal, Azure CLI, Azure PowerShell, Azure Resource Manager templates, REST API	Role information can be accessed in Azure portal, Microsoft 365 admin portal, Microsoft Graph PowerShell

Owner
Reader
Contributor
un
RG

Global Administrator
Global Reader

User Administrator
...

+ custom Roles

...
+ custom Roles



There are many built-in roles, or you can create your own custom role

Manage Azure built-in role assignments

General	
Built-in role	Description
Contributor	Grants full access to manage all resources but does not allow you to assign roles in Azure RBAC, manage assignments in Azure Blueprints, or share image galleries.
Owner	Grants full access to manage all resources, including the ability to assign roles in Azure RBAC.
Reader	View all resources but does not allow you to make any changes.
Role Based Access Control Administrator	Manage access to Azure resources by assigning roles using Azure RBAC. This role does not allow you to manage access using other ways, such as Azure Policy.
User Access Administrator	Enables you to manage user access to Azure resources.



The following is a general list of Azure built-in roles and is not an exhaustive representation.

Microsoft Entra built-in roles

Built-in role	Description
Application Administrator	Privileged role allows application registration, consent, and owner status for assigned users.
Attribute Assignment Administrator	Role allows assigning custom security attributes to Microsoft Entra objects; not included in default admin roles.
Attribute Log Administrator	Attribute Log Reader role: access audit logs for custom security attributes; not granted in default admin roles.
Authentication Administrator	Authentication Administrator role: manage authentication methods, reset passwords, and perform sensitive actions; limitations apply.
Authentication Policy Administrator	Authentication Policy Administrator: configure policies, manage credentials, tickets; limitations apply.

Global Admin



The following is a list of Microsoft Entra built-in roles and is not an exhaustive representation.

Create a Role Definition

Collection of permissions that lists the operations that can be performed

Azure

VS Code
json Lang Server

[] list
{ } object
json

Owner
Contributor
Reader
...
Backup Operator
Security Reader
User Access Administrator
Virtual Machine Contributor

Built-in

Reader Support Tickets
Virtual Machine Operator

Custom

Contributor

```
"Actions": [  
  "*"   
],  
"NotActions": [  
  "Authorization/*/Delete",  
  "Authorization/*/Write",  
  "Authorization/elevateAccess/Action"  
],  
"DataActions": [],  
"NotDataActions": [],  
"AssignableScopes": [  
  "/"  
]
```

Einschränkung

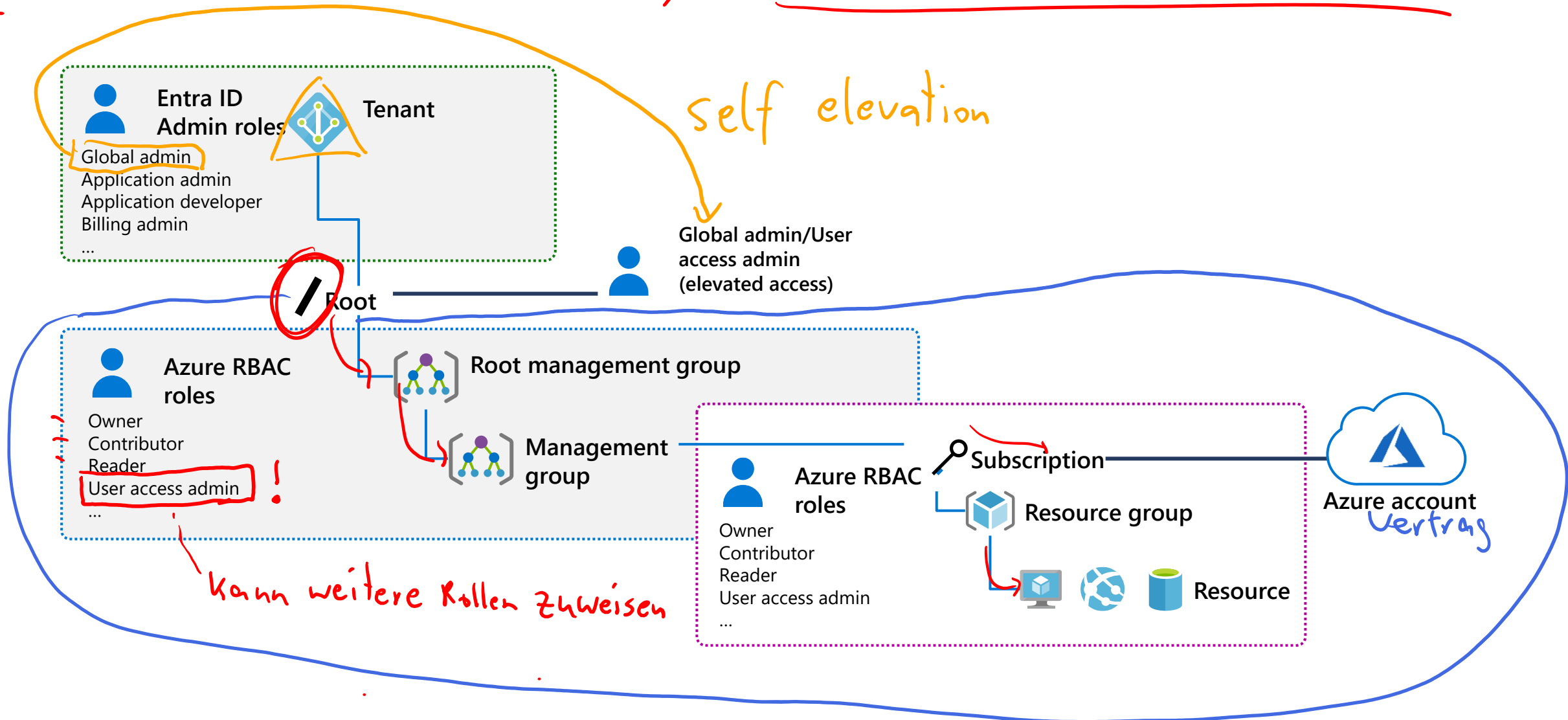
No Data Perm.

ngmt
Perm.

Data
Permission

(Apply RBAC Authentication)

Seize control of Azure



Microsoft Entra Privileged Identity Management

PIM = Rollenzuweisung auf Antrag und nur für ein Zeitfenster z.B. 2h

- PIM **manages, controls**, and **monitors access** to key resources across Microsoft services, requiring licenses.

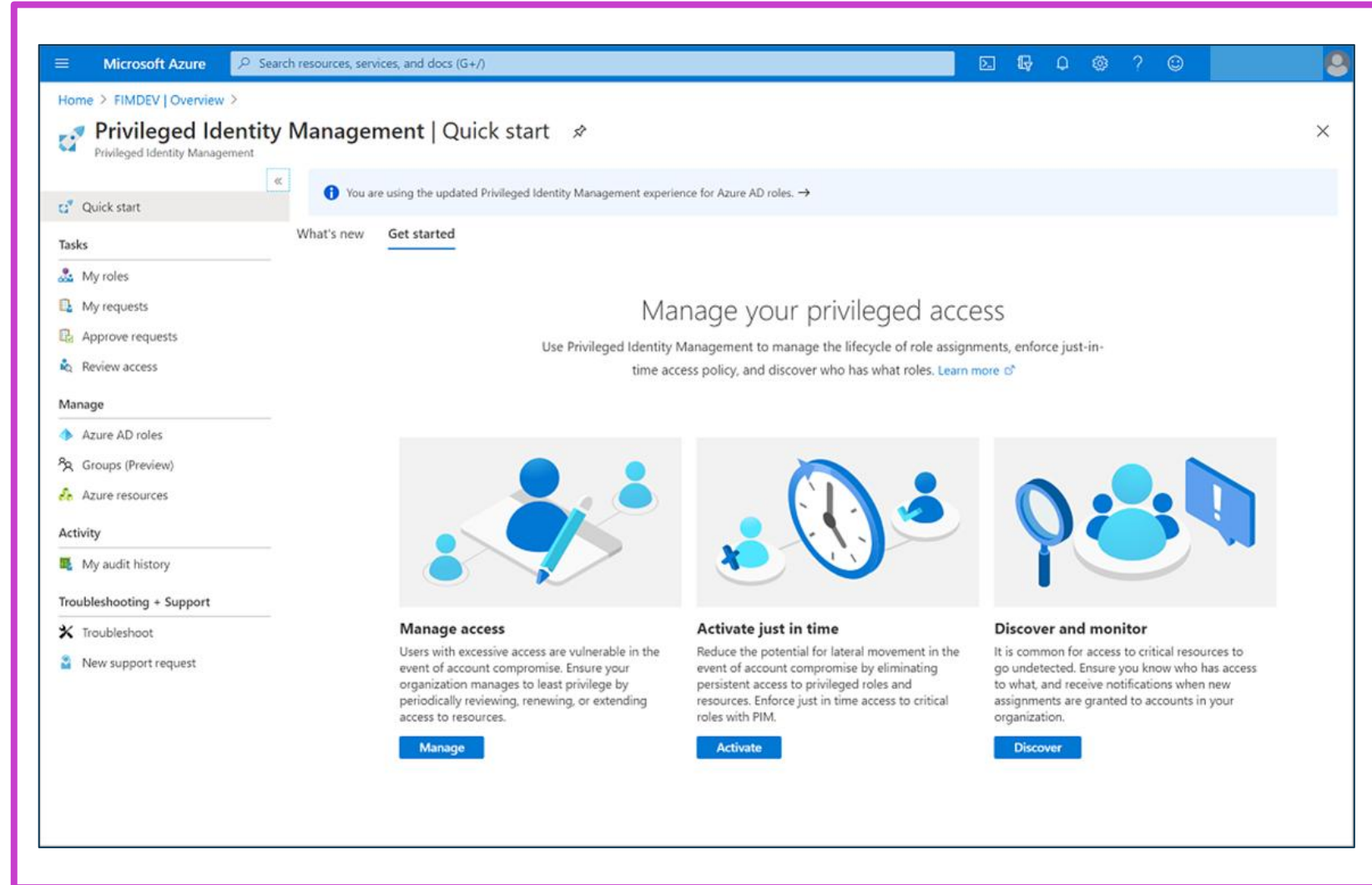
JIT

- Enables **just-in-time privileged access** and oversight for user operations in Azure and Microsoft services.

JEA

just enough admin

- Offers **role management**, **activation**, and **approval processes**, with email notifications for assignment changes.



Labs



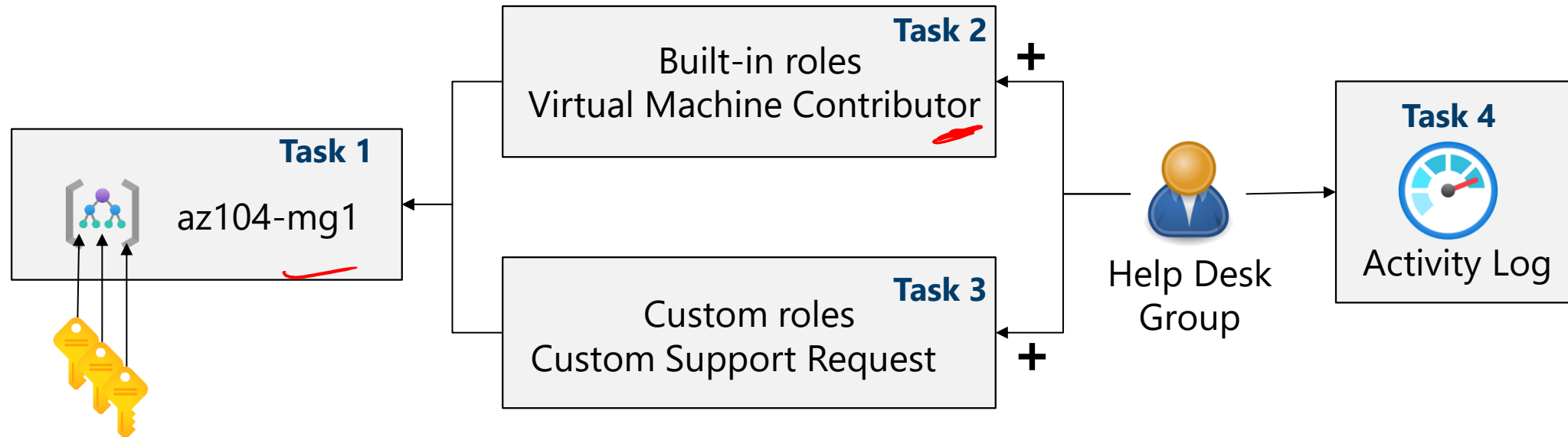
Skillable

AZ-104

Lab 02a - Manage Subscriptions and RBAC



Lab 02a – Architecture diagram



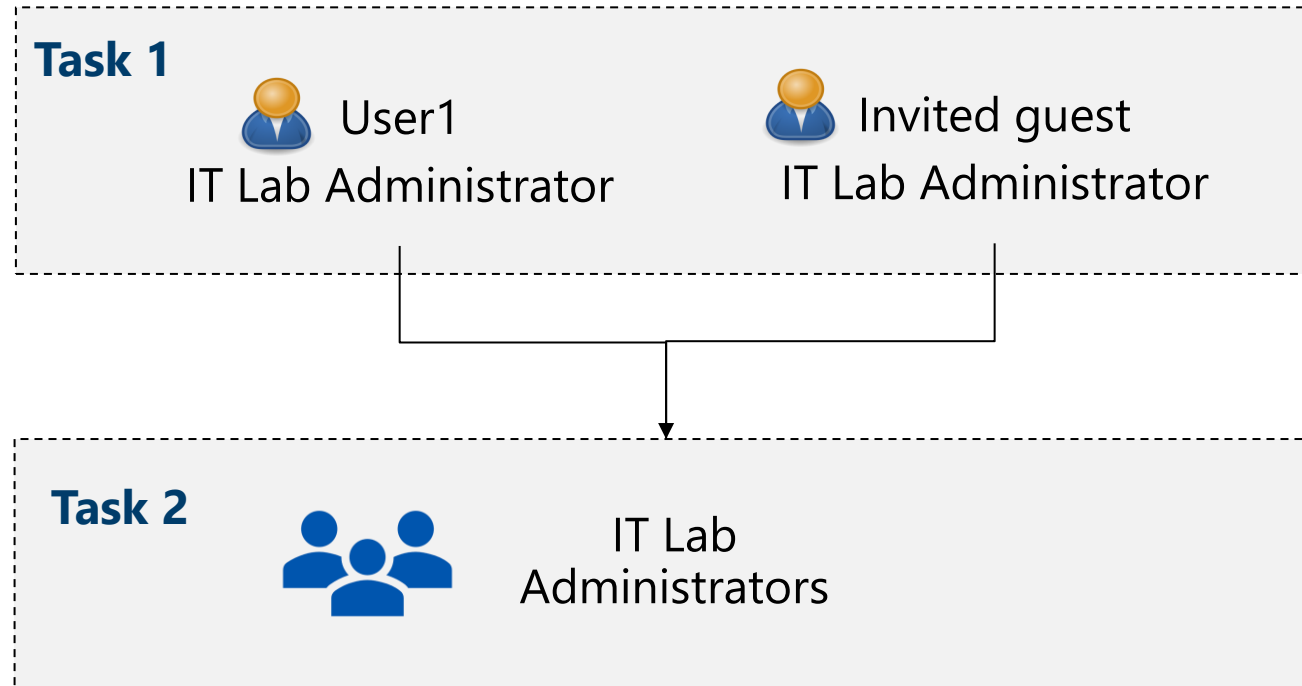
Skillable

AZ-104

Lab 01 – Manage Entra ID Identities



Lab 01 – Manage Entra ID Identities (architecture diagram)



End of presentation



